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How can we assess the success of information technologies in digital libraries? Empirical evidence from Indonesia

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ABSTRACT

This study attempts to model the factors that affect the system's use, user satisfaction, net benefits, and user loyalty to digital library services. We developed the proposed framework using Delone and McLean's information system success model. A survey instrument was adapted, validated, and assessed for reliability. The survey data were gathered from respondents (n. 2105) who are Indonesian open university (IOU) students. We evaluated the validity and reliability of the data by assessing the measurement model. We applied a structural model in the Partial Least Square Structural Equation Modelling (PLS-SEM) to report the path coefficient (β), p-value, and t-value for the main data analysis. Findings inform that all hypotheses were confirmed. Service quality had the most significant role in predicting user satisfaction, and user satisfaction was the most significant predictor of use. In addition, the net benefits variable was mostly affected by use. For user loyalty, the most robust determinant was net benefits. This research elaborates on quality criteria that can be a reference for related stakeholders to improve digital library services, satisfaction, use, net benefits, and user loyalty.

Introduction

The Internet is now the platform for many different activities that online activities have increasingly merged into people's daily lives. Internet users produce content and participate in social and interactive online activities (Mishra, 2019). In education, the digital library is one of many online systems intended to engage users' participation, in which learning resources available in digital libraries are accessible to users. Knowledge resources from digital libraries are increasingly easier to access and more effectively collected, stored, and managed. A digital library is a collection of services offered by a library that consists of digitized information files that have been catalogued, indexed, and supported by librarians to aid users (Gul & Bano, 2019). These files may contain original contents, or the outcome of pre-existing content being electronically converted, such as books, journals, papers, images, movies, etc.

The idea of libraries has been around for most of recorded history, and they are frequently considered places where information is gathered. Libraries have significantly enabled people to access and interact with various knowledge and information sources. Technology advancements have made knowledge gathering, managing, and storing much more efficient (Fakhar Manesh et al., 2021) Digital libraries gain the benefits for the development of the system, such as cutting-edge social and interactive tools, making administrators easier to create and share resources (Hussain, 2019). For a sustainable future, digital libraries must demonstrate simplification and innovation (Makarov et al., 2022) with more effective and efficient technologies supporting the system (Trace & Karadkar, 2017). To better serve users, digital libraries should regularly update and assess services (A. Habibi et al., 2022). Research on the success of digital library systems should be sustainable because their use has become more popular. Many studies have explored the success of digital libraries in education, e.g., (AbdelKader & Sayed, 2022; Alajmi & Alotaibi, 2020; Shahzad et al., 2021; Xu & Du, 2018). However, the studies were conducted before Covid-19 happened and were conducted within a medium-scaled survey in terms of study participants.

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This study, therefore, was conducted after Covid-19, in which students are accustomed to having online or distance education systems. Through a thorough literature analysis, we planned this study, followed by the decision to conduct a survey as a means of data collection that aimed to comprehend the factors affecting the system's use, user satisfaction, net benefits, and user loyalty to digital library services. Responses of 2105 IOU students were statistically reported to produce an understanding of how digital users perceive library success. This study also suggests ways to encourage more students to use digital libraries to increase and continue to use the system. Recommendations would benefit future researchers interested in similar studies and stakeholders intended to improve digital library services.

Literature review

Information technologies and library

Over the years, the fast growth of Information Technology (IT) has changed the way Library and Information Centres work, turning them into hubs of information and new ideas (Kumar, 2020). Every day, new technologies change how information is found, organised, saved, and shared, bringing libraries into a new era of being more efficient and easier to use. The digitization of material is one of the most important things that IT has brought to libraries (Martini, 2016). The change from physical to digital materials has made libraries so much bigger and more useful. People from all over the world can easily reach huge amounts of information, regardless of their location or the time of day. Users have access to a wide range of resources, such as e-books, academic journals, study papers, and multimedia content, through digital platforms and online databases. The spread of information technology has made it easier to set up digital libraries and places for people to work together. The platforms make it easy for experts, researchers, and students from different backgrounds to talk to each other and share resources. This condition creates a global community of learners for knowledge. Webinars, online workshops, and virtual conferences are now routine (Bennett & McWhorter, 2021). This makes it easier for people in the library community to keep learning and improve their skills.

Digital library systems in Indonesian open university (IOU)

Theoretically, digital library systems have been implemented in educational institutions across the globe studies (Al-Hattami, 2021; Zibak et al., 2021). In Indonesia, educational institutions have digital library systems that usually link to research databases and journals. (Maryati et al., 2020) analyzed the websites of 9 Indonesian universities' library systems in the context of designs and contents. They informed that their investigation had validated the attributes of the research support services to evaluate libraries' applications. A few libraries in Indonesia have adapted research support services such as data management, publishing of information, copyright services, financing information, and collaboration. However, there are already prospects for the services in digital libraries of a few universities, supporting instruction and research activities. The universities increasingly provide services, training for online website searches, citations and references, and access to online databases (Suwana, 2021). Over the past ten years, online libraries, including Indonesia, have been built across Asia, Africa, and Latin America (Alzahrani et al., 2019; Khan et al., 2021). The Indonesian Open University Library Service Center (OULSE) was established in 1984. The OULSE is a university library projected as a distance education library. The library materials collected are quite diverse, consisting of printed and non-printed materials. Through the internet network, users, mostly students (n. 346.584), can access the OULSE website (https://pustaka.ut.ac.id/lib/). The center has over 30 thousand titles: journal subscriptions, textbooks, references, modules, documents, research reports, theses, and dissertations.

Information systems success model

Delone and McLean developed the D&M IS success model in 1992 (Fig. 1). This IS theory aims to provide a comprehensive understanding of IS effectiveness by defining, outlining, and clarifying the links between six dimensions (net benefits, intention to use or use, satisfaction, information quality, system quality, and service quality) of success in which IS is evaluated.(DeLone & McLean, 1992) adapted Mason's (Mason, 1978) formulation of the effect level and three levels of information. (DeLone & McLean, 2003) amended and analyzed for ten years; they also published a review paper claiming that 285 publications cited their model. However, besides the many citations, critics have also addressed the weakness of the model. For example, (Seddon, 1997) claimed that the model was not a success metric and that the advantages to people and organizations should be the opposite of pleasure.

Despite the critics, many researchers used D&M IS success model in their studies (AbdelKader & Sayed, 2022; Al-Hattami, 2021; Zibak et al., 2021; Amponsah et al., 2022). For example, (Amponsah et al., 2022) assessed the success of cloud-based technology systems in finance. In education, (Sabeh et al., 2021) reported 92 studies adapted the model from 2010 to 2020, evidence that the model is essential in academia. A study for online learning in Yemen through a survey of 448 students, extending the D&M IS model with self-efficacy, informed system, information service quality, and self-efficacy significantly influenced satisfaction and use, impacting student performance (Aldholay et al., 2018). (Yakubu & Dasuki, 2018) examined the D&M IS success model for educational information systems to determine whether users in Africa intended to use the eLearning system. The results partially supported the influence of the quality antecedents on behavioral intention and user satisfaction. (Shahzad et al., 2021) used the D&M IS success model to compare males and females to E-learning portal usage. The study objective is to check the difference between male and female E-learning portals' accessibility from the students' perspective. They discovered that Information quality also supported the relationship with system use. At the same time, there is a positive relationship between user satisfaction and E-learning portals. Even though many education-based studies have adapted and extended the D&M IS success model, academics, particularly in library sciences, have paid less attention to validating the D&M success model for IS of digital library settings (Alajmi & Alotaibi, 2020; Xu & Du, 2018; Alzahrani et al., 2019).

The study

A few researchers have tried to understand user loyalty by considering how users use digital library systems, in addition to the enjoyment and net benefits. Limited scholars explored the factors affecting students' real use of digital libraries and their satisfaction regarding user loyalty, notably in the context of open universities, where prior research has mostly concentrated on several diverse themes. To address this lack of research focus, this study aims to determine what components of quality criteria or student perspectives would promote the use. Therefore, the following research question is looked at in this study: To what extent do the Delone and McLean information system success model (D&M IS success model) variables relate to the user loyalty indicator? "User loyalty" is listed as one of the dependent variables in this research, extending the IS success model (Fig. 2).

Hypotheses

System quality

System quality is the degree to which systems are "user friendly" and simple to operate (Davis, 1989), reflecting the quality of the IS. IS system quality focuses on the use of system software and hardware (Bharati & Berg, 2003). System quality includes the evaluation flexibility, functionality, easiness, relevance, integration, and quality of the IS (DeLone

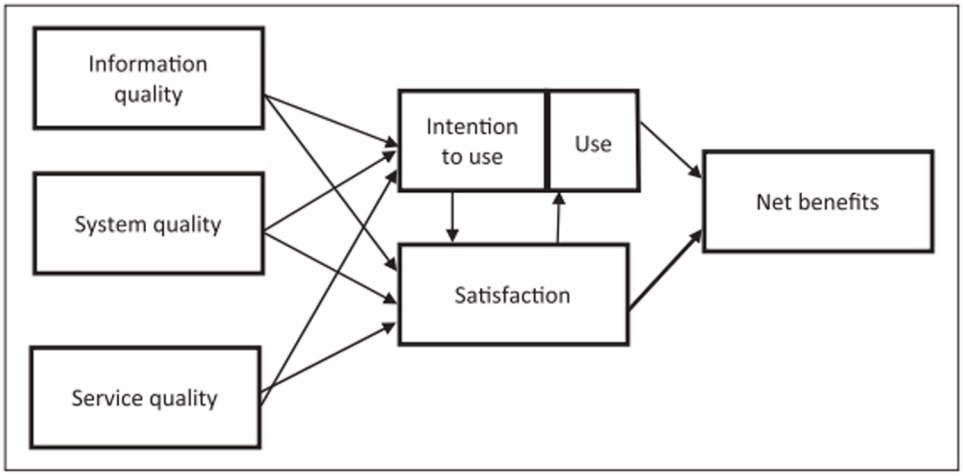


Fig. 1. D & M IS success model.

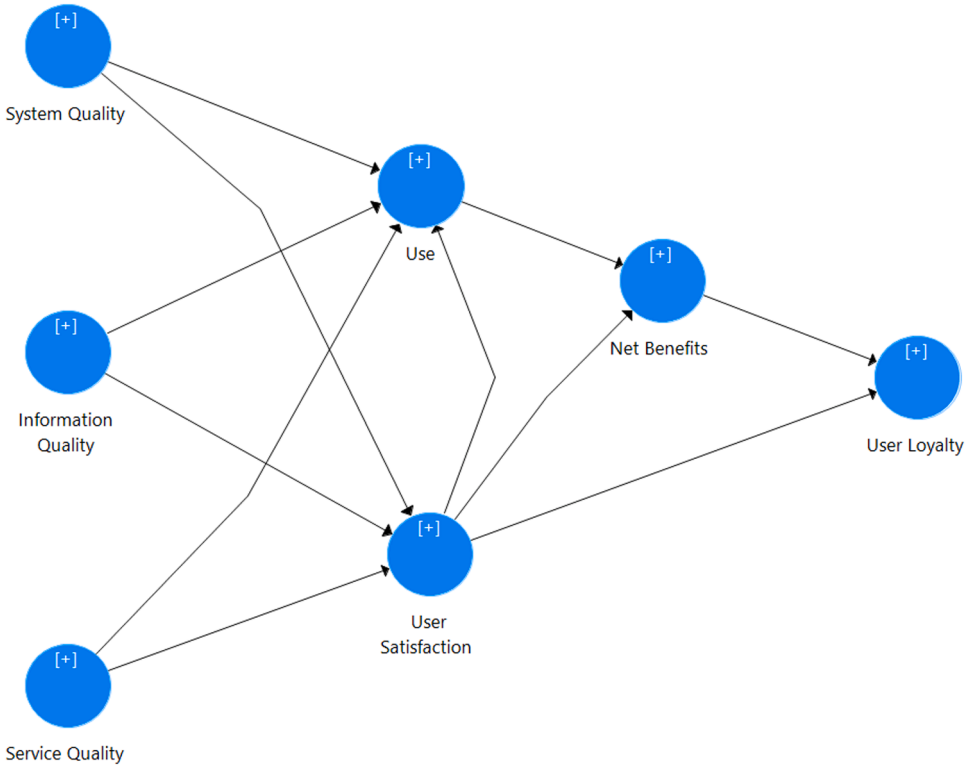


Fig. 2. Model of the study.

& McLean, 2003). In education, studies reported the key role of system quality in predicting IS use and user satisfaction (Shahzad et al., 2021; Aldholay et al., 2018; Yakubu & Dasuki, 2018). For instance, (Shahzad et al., 2021) reports showed that system quality significantly affected e-learning use ($\beta = 0.299$; $p < 0.05$) and user satisfaction ($\beta = 0.389$; $p < 0.001$) perceived by 280 Malaysian undergraduate students. In the context of the digital library, (Alzahrani et al., 2019) revealed significant influences of system quality towards use and user satisfaction. Data from 978 respondents informed system quality was the most significant determinant for intention to use digital library system ($\beta = 0.362$; $p < 0.001$); besides, it also significantly affected satisfaction ($\beta = 0.201$; $p < 0.001$). In this study, we include two hypotheses regarding the role of system quality on OULSE system use (H1) and user satisfaction (H2).

Information quality

Information quality is how the system's user perceives its outcome value. The most common criteria for assessing information quality are accuracy, thoroughness, immediacy, usefulness, and consistency. Furthermore, information quality is crucial for assessing IS efficiency (Negash et al., 2003). In the original D&M IS success model, information quality was included as a variable that correlated with use and user satisfaction (DeLone & McLean, 2003). Following the model, many studies have disclosed the significant relationships between information quality and use/user satisfaction (Shahzad et al., 2021; Aldholay et al., 2018; Yakubu & Dasuki, 2018). The IS literature for digital library systems contends that information quality significantly or insignificantly predicted use and satisfaction (Alajmi & Alotaibi, 2020; Alzahrani et al., 2019). According to (Alzahrani et al., 2019), information quality was a significant determinant of user satisfaction and behavioral

intention to use the digital library system in Malaysia. However, (Alajmi & Alotaibi, 2020) failed to report the significant relationship between system quality and digital library use. In this study, we proposed that information quality affects the use of digital library systems (H3) and user satisfaction (H4) in IOU.

Service quality

Users' subjective assessments of whether the service they get from the website meets their expectations are termed service quality (DeLone & McLean, 2003). Measuring service quality in terms of competency, follow-up service, empathetic response, assurance, dependability, and responsiveness is an effective method. Since (DeLone & McLean, 2003) employed service quality to evaluate IS success, the modified approach has generated a lot of interest in studies (AbdelKader & Sayed, 2022; Al-Hattami, 2021; Zibak et al., 2021). In the context of the study, service quality refers to users' assessment of OULSE, determining the use and user satisfaction. Few studies have informed the roles of surveys (Alajmi & Alotaibi, 2020; Shahzad et al., 2021; Xu & Du, 2018). In a recent study, (Shahzad et al., 2021) used PLS-SEM to find a significant relationship between service quality and user satisfaction and the actual use of a digital library perceived by 280 students. The findings informed that user satisfaction and actual use were significantly affected by service quality. Service was also related to satisfaction and intention in using the digital library system, revealed in a context. As a result, higher levels of service quality can result in higher levels of user satisfaction and actual use of a system. Therefore, we include two hypotheses regarding the role of service quality towards user satisfaction (H5) and the use of OULSE (H6).

User satisfaction

One of the key success factors is user satisfaction for the D&M IS model. In the initial establishment of the D&M IS success model (DeLone & McLean, 2003); they found a significant connection between system use and net benefits. User satisfaction was the most frequently cited among the six key domains of the IS success model in 1992 (DeLone & McLean, 1992). User satisfaction can be either positive or negative depending on the benefits users expect from using the IS. Studies in education have disclosed the strong relationships between user satisfaction and use/net benefits (AbdelKader & Sayed, 2022; Alajmi & Alotaibi, 2020; Shahzad et al., 2021; Xu & Du, 2018). The digital library system was also the subject of studies that reported the relationship between user satisfaction and use. With a path coefficient of 0.257, (Alzahrani et al., 2019) revealed the significant role of satisfaction and use of the digital library. (AbdelKader & Sayed, 2022) evaluated the relationship between user satisfaction and net benefit and found a significant correlation. In this study, we hypothesized the relationship between user satisfaction and use (H7) and between user satisfaction and net benefits (H8).

In addition to the role of user satisfaction on use and net benefits, we also report its effect on user loyalty in using OULSE perceived by the users, the students of IOU. Loyalty is a feeling of adoration for an organization's personnel, products, or services. Using this idea, we describe user loyalty as an emotional connection to digital library systems' information resources and support services. By consistently using the services and systems, users can continue to be devoted to digital libraries. Prior studies have looked at factors influencing patron loyalty in academic and public libraries, such as collection quality, perceived value, customer satisfaction, facilities, the effects of services, and accessibility (Bae & Cha, 2015; Gede Mahatma Yuda Bakti & Sumaedi, 2013; Keshvari et al., 2015). In brief, the human qualities of user services had the greatest impact on customer loyalty. Thus, customer satisfaction has an impact on user loyalty (H9).

Use

The amount of technology infrastructure is the main focus of these systems' use goals. To figure out how much the system was actually used in a voluntary setting, the total number of functions used, how often it was used, and how long it was used were all recorded objectively. (Davis, 1989) began a subjective evaluation of the intention to use the system using a questionnaire to ascertain utilization. The unified theory of acceptance and use of technology (UTAUT) paradigm states that a user's adoption of technology is based on how they utilize it. In this study, the use of a digital library system of voluntary use may be a suitable success metric (A. Habibi et al., 2022). The major goal of this study was to determine the relationship between use and net benefits. Many studies in IS maintain that use is the best predictor of net benefits (AbdelKader & Sayed, 2022; Shahzad et al., 2021; Rokhman et al., 2022). (AbdelKader & Sayed, 2022) discussed a significant relationship between use and net benefits regarding the Egyptian Knowledge Bank (EKB) as the biggest digital library in Egypt. However, use was not significantly affected by e-learning use during Covid-19 (Rokhman et al., 2022). H10 represents the relationship between use and net benefits.

Net benefits

In the original model, the net benefits dimension is defined as an innovative variable that united two impact dimensions (organizational and individual) into one. This overall dependent variable of the D&M IS Success Model refers to the degree to which IS contributes to the success of individuals, groups, organizations, industries, and nations. In this study context, net benefits play both as dependent and independent variables. For the later, one hypothesis is provided; net benefits are hypothesized to significantly predict user loyalty (H11).

User loyalty

This study extended the model by adding one variable, user loyalty, to be predicted by net benefits. User loyalty refers to a feeling of adoration for an organization's personnel, products, or services (Twum et al., 2022; Soltani et al., 2018). We chose loyalty to understand the sentiment of admiration for digital library users, IOU students, to use the products and services IOU digital library in the context of this research.

Methodology

Design

All the constructs and the corresponding measurement items were adapted from previous literature to satisfy content validity (DeLone & McLean, 2003; Gefen, 2002). A survey was used since it efficiently collects information from a large number of respondents, allowing for diverse perspectives and results that can be generalized. Quantitative data for statistical analysis are cost-effective relative to other methods and provide anonymity and privacy. It is also accessible via multiple platforms, allowing them to reach diverse populations (Al-Adwan et al., 2022).

The present study measured all survey indicators with a five-point scale of Likert (from 1 = strongly disagree to 5 = strongly agree). Two sections of the instrument were established. Demographic information was established, including educational level, gender, age, user experience, and frequency of use of digital libraries. Theoretical constructs of the model were included in the second section of the instrument (Attachment A). As we conducted the survey in Indonesia, the questionnaire was translated into Indonesian by an expert in teaching English as a foreign language and translated back into English for consistency. We also gained opportunities to interview some of them to cross-check the data. We piloted the instrument to 30 users (IOU students) to check the reliability through Cronbach's alpha. From the pilot study and interview feedback, we revised some items to improve the clarity and understandability of the questionnaire.

Data collection

IOU has more than 400 thousand students from all over Indonesia;

most utilize digital libraries for learning. For the data collection of the current study, we used Google Forms by inviting students of IOU through social media direct messages, calls, and emails to fill in the survey. Efforts were made to ensure that students from diverse majors, such as education, economics, library science, library science, and law. Within the process, we initially informed the purpose of the study, discussing all variables in the instrument and the types of responses. We asked the respondents to complete the questionnaire based on their experiences and knowledge of using digital libraries in their learning. Responses submitted were in the same score given to all items, and weak or limited responses were deemed invalid and removed from the analysis (Habibi et al., 2020); 127 responses were deleted due to the said reasons. In total, 2105 valid responses were collected and used for data analysis. In detail, Table 1 exhibits the respondents' demographic information (gender, age, information, educational level, and Digital library use frequency)

Analysis and findings

The reliability and validity of the measurement model were evaluated and reported through the reports of loadings, internal consistency reliability (ICR), convergent validity, and discriminant validity (Habibi et al., 2021; Ogbeibu et al., 2018). We used PLS-SEM analysis to test the research hypotheses with a two-step approach, measurement and structural models (Alfalah, 2023; Habibi et al., 2023; Jadil et al., 2022). The current SmartPLS was selected due to its effectiveness and efficiency (AbdelKader & Sayed, 2022; Mohammed et al., 2023) (Table 2).

Findings

Measurement model

A measurement model examines a model's construct validity and reliability (A. Habibi et al., 2022). We detailed the steps for the reports of loadings, ICR, convergent validity, and discriminant validity. Items should be valued at more than 0.500 for loadings (Ogbeibu et al., 2018; bt A. Hamid Shokery et al., 2016). An item (SERQ1) was dropped since the loading was below the threshold. The computation resulted in all loading values for the measurement model in this study being more than 0.500 (Table 4). Following the loading assessment, ICR was evaluated, described as an effort to evaluate the consistency of computation across items; the computation should be from 0 to 1; the greater the value, the higher the reliability. Cronbach's alpha and Rho_A should be from 0.700 to 0.950. The results, displayed in Table 3 and Fig. 3, verify that all results comply with the standards; all measures for ICR score are greater than.700. Service quality received the lowest Cronbach's alpha (0.863),

Table 1
Demography information (n. 2105).

Type	n	%
Gender		
Female	1393	66.18
Male	712	33.83
Age		
<20	299	14.20
20–30	1021	48.50
>30	785	37.29
Education level		
Diploma	43	2.04
Undergraduate	1925	91.45
Master	113	5.37
Doctoral	24	1.14
Digital library use frequency		
Never	0	0
Seldom	786	37.34
Sometimes	906	43.04
Often	342	16.25
Frequently	71	3.37

Table 2
α, Rho_A, CR and AVE.

Variable	α	rho_A	AVE
Information Quality	0.903	0.903	0.837
Net Benefits	0.896	0.897	0.828
Service Quality	0.863	0.865	0.880
System Quality	0.895	0.896	0.827
Use	0.938	0.938	0.843
User Loyalty	0.914	0.914	0.853
User Satisfaction	0.949	0.949	0.907

while user satisfaction received the best score (0.949). Similarly, Rho_A values were also satisfactory ranging from 0.865 (service quality) to 0.949 (user satisfaction). Convergent validity, a component of construct validity, stipulates that testing with the same or similar constructs ought to have a high degree of correlation; we used average extracted variance (AVE) as the measurement; AVE values must be > 0.500 or 50% of the indicators' variance (A. Habibi et al., 2022). All variables had good AVE values of more than 0.500, ranging from 0.827 (system quality) to 0.907 (user satisfaction).

If two constructs are supposed to be unconnected, discriminant validity tests their relationship. We developed the discriminant validity by using the heterotrait-monotrait (HTMT) ratio of correlations as the main indicator. The HTMT cut-off value for this investigation was.900. When the value exceeds.900, discriminant validity becomes a problem. Consequently, the values must be lower than 0.900. Some items must be drooped for HTMT to be lower than 0.900. In the first calculation, some HTMT values were higher than 0.900. We dropped the items one by one to establish the discriminant validity; the dropped items were SERQ4, NB4, and NB5. After the deletion process, all the HTMT values were lower than 0.900, ranging from 0.725 to 0.898. Cross-loading assessment was also used to support the evidence of the discriminant validity establishment. The cross-loadings assess items that should have higher values on the same construct than other constructs. Table 4 shows that all cross-loading values emerge, establishing discriminant validity of the prosed model. The model was validated based on the measurement model assessments (loadings, ICR, convergent and discriminant validity).

Structural model

The hypotheses established were examined by conducting a bootstrapping analysis with 5000 resamples (Hair et al., 2022). The statistical findings of the study are exhibited in Table 5 and Fig. 4. As predicted, the quality factors were positively correlated to user satisfaction, namely system quality ($\beta = 0.289$), information quality ($\beta = 0.297$, $p < 0.01$), and service quality ($\beta = 0.335$), which confirm H2, H4, and H6. The correlations between all independent variables to use were also significant; system quality ($\beta = 0.097$), information quality ($\beta = 0.108$), service quality ($\beta = 0.180$), and user satisfaction ($\beta = 0.528$), confirming H1, H3, H5, and H7. The bootstrapping procedure also tested the key role of user satisfaction and use in affecting net benefits. Significant positive correlations between user satisfaction ($\beta = 0.430$), use ($\beta = 0.460$), and net benefits support H8 and H10. For user royalty, two determinants were also reported to be significant. User satisfaction significantly affected user royalty ($\beta = 0.359$) that supports H9; net benefits were substantial in affecting user royalty ($\beta = 0.478$) that supports H9.

In addition to the assessment of beta, t-value, and p-value, the effect size (f^2) of each relationship were also computed. Effect size computations in statistics aim to measure the strength of the correlations between 2 constructs as the parameter for a hypothetical preparation (Hair et al., 2022). All relationships between the constructs in this current study show the existence of effect size (Table 5). The highest effect size value emerged regarding the relationship between user satisfaction and use ($f^2 = 0.284$). Meanwhile, the lowest value was the relationship

Table 3
HTMT.

	Information Quality	Net Benefits	Service Quality	System Quality	Use	User Loyalty
Net Benefits	0.825					
Service Quality	0.898	0.818				
System Quality	0.861	0.773	0.832			
Use	0.805	0.889	0.832	0.781		
User Loyalty	0.725	0.849	0.752	0.698	0.809	
User Satisfaction	0.849	0.879	0.864	0.829	0.877	0.801

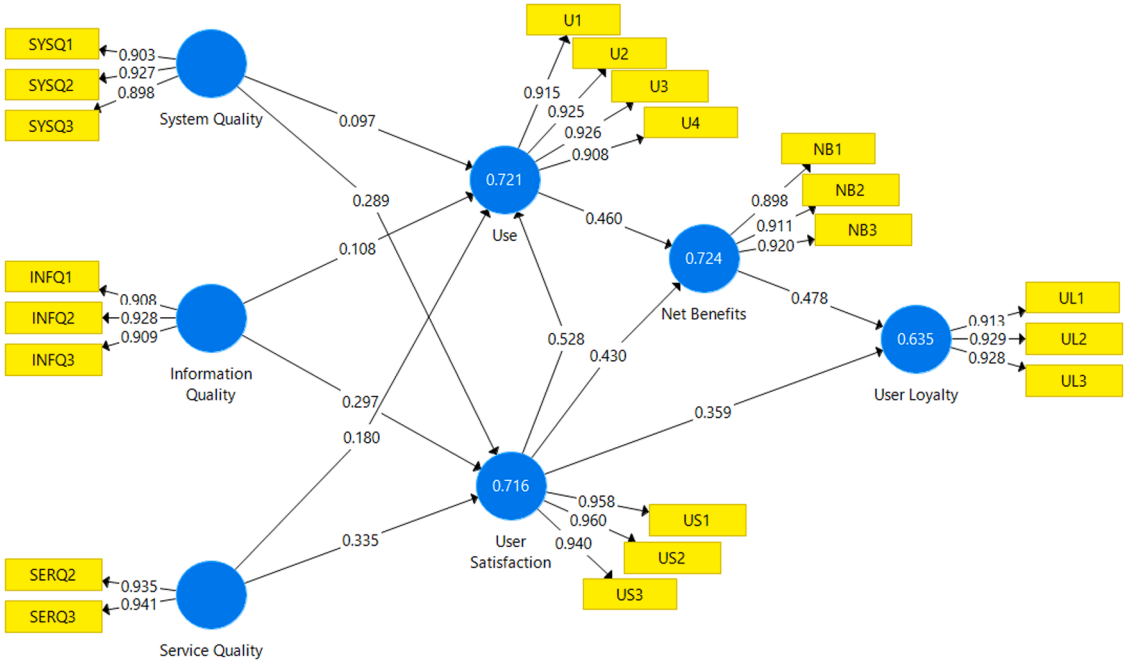


Fig. 3. Measurement model of the study.

Table 4
Loading and cross loading.

	Information Quality	Net Benefits	Service Quality	System Quality	Use	User Loyalty	User Satisfaction
INFQ1	0.908	0.663	0.710	0.713	0.665	0.594	0.715
INFQ2	0.928	0.692	0.726	0.709	0.686	0.609	0.726
INFQ3	0.909	0.684	0.740	0.702	0.682	0.605	0.717
NB1	0.648	0.898	0.617	0.608	0.709	0.672	0.705
NB2	0.678	0.911	0.670	0.627	0.755	0.709	0.742
NB3	0.699	0.920	0.676	0.654	0.762	0.716	0.764
SERQ2	0.736	0.659	0.935	0.669	0.684	0.604	0.715
SERQ3	0.751	0.692	0.941	0.704	0.721	0.650	0.751
SYSQ1	0.707	0.630	0.641	0.903	0.649	0.569	0.678
SYSQ2	0.719	0.643	0.686	0.927	0.662	0.597	0.717
SYSQ3	0.684	0.616	0.670	0.898	0.641	0.555	0.688
U1	0.671	0.717	0.690	0.658	0.915	0.676	0.754
U2	0.694	0.767	0.702	0.668	0.925	0.686	0.759
U3	0.674	0.744	0.691	0.655	0.926	0.678	0.761
U4	0.683	0.768	0.669	0.648	0.908	0.711	0.767
UL1	0.603	0.711	0.594	0.573	0.690	0.913	0.671
UL2	0.618	0.710	0.635	0.588	0.686	0.929	0.702
UL3	0.604	0.710	0.624	0.587	0.700	0.928	0.695
US1	0.746	0.764	0.747	0.726	0.788	0.707	0.958
US2	0.752	0.765	0.751	0.734	0.782	0.713	0.960
US3	0.749	0.788	0.737	0.723	0.795	0.714	0.940

between system quality and use ($f^2 = 0.011$).

Discussion

The current study aimed to evaluate the extension of IS Success model to elaborate on the factors influencing user satisfaction, use, net benefits, and user loyalty of the digital library in the Indonesian open university (IOU), understanding information, system, and service quality and their impact on user satisfaction and use. Both use and user satisfaction were measured for their influence on net benefits. Finally,

Table 5
Structural model.

Hypothesis	Path	β	t value	p values	f^2
H1	System quality -> use	0.097	3.677	0.000	0.011
H2	System quality -> user satisfaction	0.289	10.127	0.000	0.107
H3	Information quality -> use	0.108	3.604	0.000	0.011
H4	Information quality -> user satisfaction	0.297	10.184	0.000	0.091
H5	Service quality -> use	0.180	6.722	0.000	0.034
H6	Service quality -> user satisfaction	0.335	12.318	0.000	0.133
H7	User satisfaction -> use	0.528	18.054	0.000	0.284
H8	User satisfaction -> net benefits	0.430	15.668	0.000	0.211
H9	User satisfaction -> user loyalty	0.359	11.889	0.000	0.121
H10	Use -> net benefits	0.460	16.778	0.000	0.242
H11	Net benefits -> user loyalty	0.478	15.860	0.000	0.214

user loyalty was hypothesized to be predicted by user satisfaction and net benefits.

The empirical data show that hypotheses (1 to 6) emphasize that three predictors of the proposed model (information, system, and service quality) significantly correlated to digital library user satisfaction and use. All three quality measures strongly influenced user satisfaction and use of the system of IOU. It is certain that if stakeholders, OULSE, keep updating the digital library for its services and functions of knowledge and information sources, users will gain more satisfaction or contentment. Students might have more intention to come to check the digital library for newly available sources of information and knowledge, such as published books and articles, as well as other reading materials. When the three types of user expectations on OULSE improve, satisfaction and daily use will also be significant. Consistent with the findings of (Shahzad et al., 2021; Aldholay et al., 2018; Yakubu & Dasuki, 2018), the impact of information quality and system quality on users' satisfaction and usage was observed. The findings of this study provide valuable insights into the factors that influence user satisfaction and use of digital libraries, suggesting that information quality, system quality, and service quality are key predictors of user satisfaction and use of the digital library of IOU. These results highlight the importance of ensuring that digital libraries are regularly updated with new sources of information and knowledge and that the services and functions of the library are constantly improved to meet user expectations. Universities should prioritize the needs of their users and regularly seek feedback to identify areas of improvement. Universities should also consider adopting open-source library systems, as this may help to improve the quality of their systems, services, and information sources.

In addition, the roles of user satisfaction towards use(H7), net benefits (H8), and user loyalty (H9) are significantly reported by this study. This is evidence that stakeholders should consider the importance of ensuring the quality of the systems to improve the satisfaction of the use, net benefits, and user loyalty of the digital library. In this case, (Shahzad et al., 2021) also recommend that librarians should consider user satisfaction, which leads to the system use triggered by the net benefits and loyalty perceived by the users of OULSE of the various functionalities and support services integration. This finding demonstrates that user satisfaction can be considered a predictor of use. Earlier research utilized user satisfaction as the predictor of use and net benefits has also informed significant relationships (AbdelKader & Sayed, 2022; Alajmi & Alotaibi, 2020; Shahzad et al., 2021; Xu & Du, 2018). Limited studies explored the correlations between user satisfaction and user loyalty, and few elaborated on the significant relationship between the two (Bae & Cha, 2015; Gede Mahatma Yuda Bakti & Sumaedi, 2013; Keshvari et al., 2015).

Use was reported to have a significant relationship with net benefits, confirming the H10 of the current study. These findings emphasize that the increase in both system use of OULSE perceived by the users would lead to more benefits to the users, IOU students. This aligns with the previous studies (AbdelKader & Sayed, 2022; Shahzad et al., 2021; Rokhman et al., 2022). For example, (AbdelKader & Sayed, 2022)

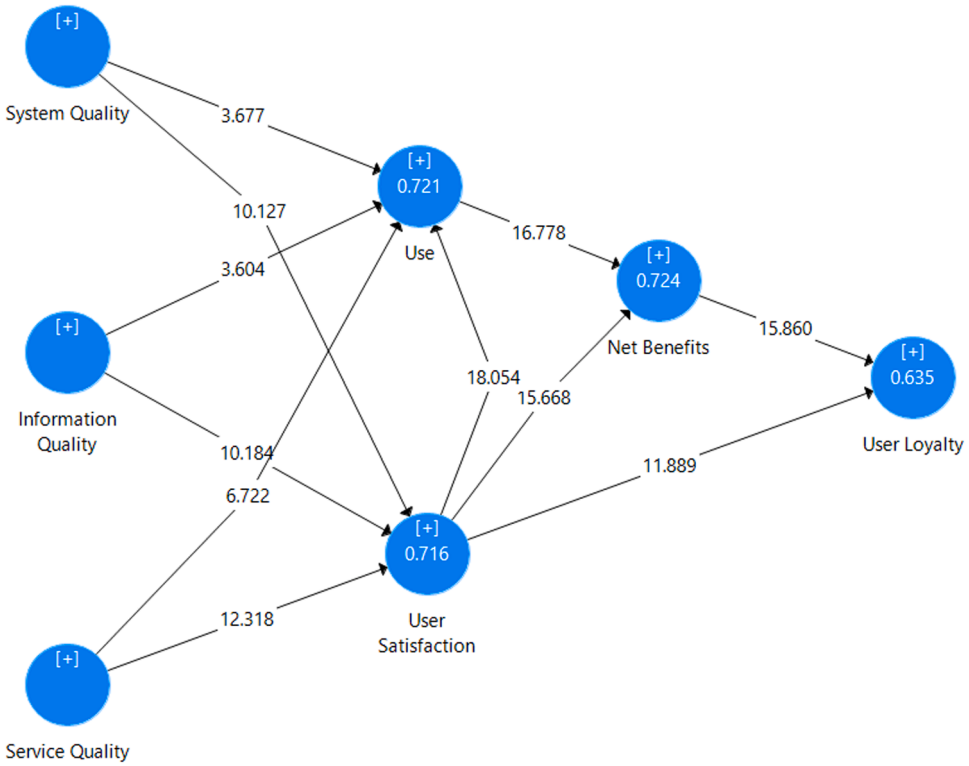


Fig. 4. Final model.

discussed a significant relationship between use and net benefits regarding the EKB as Egypt's most prominent digital library. Lastly, net benefits were an important determinant of user loyalty regarding OULSE. Limited prior studies explored this relationship. This study recommends that librarians and service providers like OULSE should take everything possible to improve the net benefits of digital libraries to improve users' loyalty so that users are willing to utilize the services continually.

Contributions to literature

This study sheds some light on research into how people use digital libraries at IOU. Some limitations emerged from this study. This study only looked at IOU students' perceptions, so it cannot be used to generalize about people from other countries or cultures. Self-reported survey questionnaires have some problems that should be considered. More research with a qualitative or mixed-method approach is needed to get a full picture of users' intentions and satisfaction. This research informs quality criteria and the effect of satisfaction, use, net benefits, and users' loyalty. The research model is supported by real-world evidence in all suggested directions. So, this research has important consequences for both theory and practice. This research adds to what is known about e-library implementation projects. This research informs that people with a stake in digital library services should consider how happy users or students are by improving the quality of information, systems, and services.

Implications for practice

The results give management ideas for improving digital libraries' real-life use. First, when the quality of the information, the system, and the service is higher, users are happier and more likely to use digital library systems. So, the people who make the system need to pay attention to how easy it is to use, how well it works, and how it can help people. Service and information quality are two quality criteria that greatly affect how satisfied students are with the system's use. Providers should give students up-to-date information about online resources and various service options. Second, the people in charge need to pay attention to how happy students are with the system and how likely they are to use it. The results of this study show that stakeholders in digital library systems should focus on the quality of the system, the information, and the services if they want to make students happier and more likely to use them. It raises the question of measuring quality in three ways instead of just one overall way.

A student satisfaction survey should be done regularly to get feedback and find out what they want and how happy they are. Students might think the system's quality is better if the server bandwidth for the digital library system is constantly improved and if information and tasks are tailored to each student. A knowledgeable librarian or smart web agent can solve technical problems quickly, and a 24/7 online service desk can help students feel less frustrated, reducing the number of complaints from students and encouraging them to use the digital library system more in the future. Third, providers of digital library systems can add features from sub-social networks like Google+ and Facebook, as well as plug-ins that let students know when new services or resources are available.

Conclusion

The goal of this study was to evaluate the extension of the IS Success model to explain the factors that affect user satisfaction, use, net benefits, and user loyalty of the digital library in the Indonesian open university (IOU), as well as how information, system, and service quality affects user satisfaction and use. Use and satisfaction were reported to understand their effect on net benefits. Lastly, it was hypothesized that the satisfaction and net benefits could be used to predict how loyal a user would be. The measurement model tests (loadings, ICR, convergent and discriminant validity) showed that the model was valid and reliable. All the hypotheses are valid, shown by the evidence that user satisfaction and use are linked to the quality of information, systems, and services. The results of this study give important information about what makes users happy and why they use IOU digital libraries or the systems. They suggest that the quality of the information, the quality of the system, and the quality of the service are essential for the satisfaction and use of the IOU digital library. By keeping high standards for the quality of the information, the quality of the system, and the quality of the service, a digital library system could show the commitment to meeting the needs of its users and creates an environment where users feel valued, respected, and encouraged to explore the vast amount of knowledge at their fingertips. Focusing on these key points not only makes users happier, but also helps the digital library stay relevant and successful in a digital world that is becoming more competitive and always changing. The roles of user happiness with use and net benefits are also important for user loyalty. This is proof that those with a stake in the digital library and the system should consider how important it is to ensure user happiness, net benefits, and loyalty. This study provides theoretical and practical recommendations to support the elaboration of the findings' discussions.

Attachment A

Demographic	Age	Usia
	Education level	Tingkat Pendidikan
	Gender	Jenis kelamin
	Using experience	Menggunakan pengalaman
	Frequency of use	Frekuensi penggunaan
System quality (SYSQ)	SYSQ1: The digital library in our university is reliable.	SYSQ1: Perpustakaan digital di universitas kami bisa diandalkan.
	SYSQ2: The navigation of this digital library is effective.	SYSQ2: Navigasi website perpustakaan digital efektif.
	SYSQ3: The layout of this digital library is clear.	SYSQ3: Tata letak (layout) website perpustakaan digital jelas.
Information quality (INFQ)	INFQ1: The information provided by this digital library is up to date.	INFQ1: Informasi yang disediakan oleh perpustakaan digital mutakhir.
	INFQ2: The information provided by this digital library is accurate.	INFQ2: Informasi yang disediakan oleh perpustakaan digital akurat.
	INFQ3: The information provided by this digital library is comprehensive.	INFQ3: Informasi yang disediakan oleh perpustakaan digital lengkap
Service quality (SERQ)	SERQ1: The digital library in our university provides on-time services.	SERQ1: Perpustakaan digital menyediakan layanan tepat waktu.
	SERQ2: The digital library in our university provides prompt responses to my questions.	SERQ2: Perpustakaan digital memberikan tanggapan cepat atas pertanyaan saya.
	SERQ3: The digital library in our university provides personalized services.	SERQ3: Perpustakaan digital menyediakan layanan yang dipersonalisasi.
	SERQ4: The digital library in our university provides professional services.	SERQ4: Perpustakaan digital menyediakan layanan profesional.
Use (U)	U1: Using the digital library enables me to accomplish tasks more quickly.	U1: Menggunakan perpustakaan digital memungkinkan saya menyelesaikan tugas lebih cepat
	U2: Using the digital library has improved my job performance.	U2: Menggunakan perpustakaan digital meningkatkan kinerja belajar
	U3: Using the digital library has made my tasks easier.	U3: Menggunakan perpustakaan digital membuat tugas saya lebih mudah
	U4: I find the digital library useful in my study.	U4: Saya pikir perpustakaan digital berguna dalam studi saya

(*continued on next page*)

(*continued*)

User satisfaction (US)	US1: I feel satisfied with the services provided by this digital library. US2: I feel contented with the services provided by this digital library. US3: I like the services provided by this digital library.	US1: Saya merasa puas dengan layanan yang diberikan oleh perpustakaan digital. US2: Saya merasa puas dengan layanan yang diberikan oleh perpustakaan digital. US3: Saya menyukai layanan yang disediakan oleh perpustakaan digital.
Net benefits (NB)	NB1: The digital library help overcome the limitations of the traditional library NB2: Using the digital library causes an improvement in my reading literacy NB3: The digital library facilitates easy access to literature NB4: The digital library enhances communication among students NB5: Digital library use causes improved decision-making in learning	NB1: Perpustakaan digital membantu mengatasi keterbatasan perpustakaan tradisional NB2: Menggunakan perpustakaan digital meningkatkan literasi membaca saya NB3: Perpustakaan digital memfasilitasi akses mudah ke sumber bacaan/ literatur. NB4: Perpustakaan digital meningkatkan komunikasi dengan petugas perpustakaan. NB5: Penggunaan perpustakaan digital meningkatkan pengambilan keputusan dalam pembelajaran
Use loyalty	UL1: I intend to use this digital library in the future. UL2: I will invite my classmates to use this digital library. UL3: I will continue using this digital library rather than any alternative tools in the future	UL1: Saya berniat menggunakan perpustakaan digital ini di masa depan. UL2: Saya akan mengajak teman sekelas saya untuk menggunakan perpustakaan digital. UL3: Saya akan terus menggunakan perpustakaan digital.

Declaration of Competing Interest

All authors declare that there are no financial conflicts of interest to disclose.

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